MEDICAL INSURANCE STATISTICAL MODELING DASHBOARD

Lab 4: Applied Statistical Modeling

Course: Statistical Modeling with Python
Course Code: DS602
Programme: M.Sc. Data Science
Semester: 1
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1. Project Overview

The Medical Insurance Statistical Modeling Dashboard is an interactive Streamlit application developed for statistical analysis of medical insurance charges.

The project combines exploratory data analysis, hypothesis testing, multiple linear regression, residual diagnostics, live prediction, and web deployment.

The purpose of the project is to understand how demographic and lifestyle variables influence medical insurance charges and to present the statistical results through an interactive dashboard.

Dashboard: https://medical-insurance-statistical-modeling-dashboard.streamlit.app/

Dataset source: https://www.kaggle.com/datasets/apoorvayerpude/healthcare-dataset/data

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2. Project Objectives

1. To explore the medical insurance dataset using descriptive statistics and visualizations.
2. To compare insurance charges between two groups.
3. To test whether insurance charges differ across geographic regions.
4. To build a multiple linear regression model using Ordinary Least Squares.

5. To interpret regression coefficients, p-values, confidence intervals, and model fit.
6. To check important regression assumptions.
7. To calculate prediction and confidence intervals for new individuals.
8. To develop an interactive Streamlit dashboard.
9. To deploy the application using Streamlit Community Cloud.

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3. Dataset Description

The project uses the Medical Insurance Costs dataset, containing information about individuals and their medical insurance charges. The dataset has 1,338 observations and 7 variables.

Variable	Description	Data Type
age	Age of the individual	Numerical
sex	Sex of the individual	Categorical
bmi	Body Mass Index	Numerical
children	Number of dependent children	Numerical
smoker	Smoking status	Categorical
region	Residential region	Categorical
charges	Medical insurance charges	Numerical

Dependent variable: `charges`
Independent variables: `age, bmi, children, sex, smoker, region`

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4. Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Statsmodels, Streamlit, Git, GitHub, Streamlit Community Cloud, Visual Studio Code

Purpose of Each Library

Library	Purpose
Pandas	Reading, filtering, transforming, and summarizing the dataset.
NumPy	Numerical operations.
Matplotlib	Creating statistical plots.
Seaborn	Improved visualizations such as histograms, scatter plots, and heatmaps.
SciPy	Hypothesis tests including Shapiro-Wilk, Levene's test, Mann-Whitney U test, and ANOVA.
Statsmodels	Fitting the OLS regression model, confidence intervals, Q-Q plots, and regression diagnostics.
Streamlit	Creating the interactive web dashboard.

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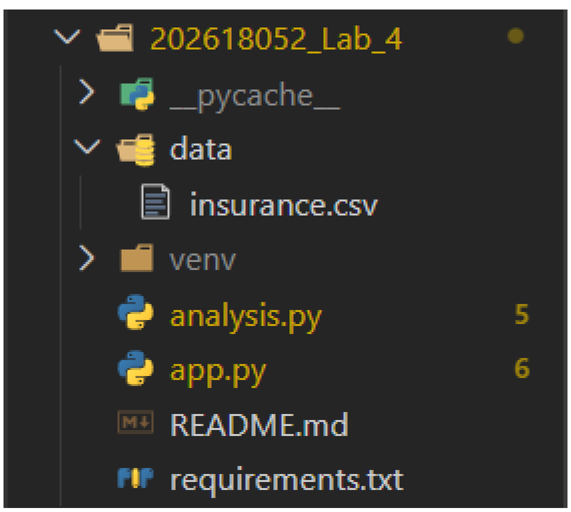
5. Project Structure



app.py — The main Streamlit dashboard file. It contains the interactive interface, filters, statistical tests, regression model, predictions, and diagnostic plots.

analysis.py — The standalone analysis file. It performs the statistical analysis through console output and Matplotlib figures.

requirements.txt — Contains all Python packages required to run the project.

data/insurance.csv — Contains the medical insurance dataset.

README.md — Contains project documentation, methodology, findings, and deployment instructions.

The **venv** folder is a local development environment and is excluded from version control. It is used only to install and run the required packages locally and is not part of the GitHub repository or the Streamlit Cloud deployment.

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6. Data Loading

The dataset is loaded using Pandas:

```
df = pd.read_csv(
    Path(__file__).parent / "data" / "insurance.csv"
)
```

The `Path` object is used to create a reliable file path relative to the Python file. This is better than using only:

```
pd.read_csv("data/insurance.csv")
```

because the application can then locate the dataset even if it is launched from a different working directory.

In the Streamlit application, the data-loading function uses:

```
@st.cache_data
```

This improves performance because Streamlit does not need to load the dataset again during every interaction.

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7. Initial Data Inspection

The project performs the following initial checks:

- Displays the first five rows.
- Checks the number of observations and variables.
- Displays column names.
- Checks data types.

- Checks missing values.
- Checks duplicate records.

Check	Result
Number of observations	1,338
Number of variables	7
Missing values	0
Duplicate rows	1

One duplicate row was identified during inspection. It was retained because the analysis was performed on the original dataset.

The dataset is therefore suitable for statistical analysis after basic inspection.

PART A: EXPLORATORY DATA ANALYSIS

8. Descriptive Statistics

The following numerical variables are analyzed: age, bmi, children, charges.

The project calculates: Mean, Median, Standard deviation, First quartile, Third quartile, Interquartile range, Skewness, Kurtosis.

Mean — The mean is the arithmetic average of all observations.

Mean = Sum of all observations / Number of observations

Median — The median is the middle value after arranging all observations in ascending order. The median is useful when the data contains extreme values because it is less affected by outliers than the mean.

Standard Deviation — Standard deviation measures the amount of variation or dispersion in the data. A large standard deviation indicates that the observations are spread widely around the mean.

Interquartile Range — The interquartile range measures the spread of the middle 50 percent of the data.

IQR = Q3 - Q1

Where Q1 is the 25th percentile and Q3 is the 75th percentile.

Skewness — Skewness measures the asymmetry of a distribution.

- Positive skewness means the distribution has a longer right tail.
- Negative skewness means the distribution has a longer left tail.
- Skewness close to zero indicates approximate symmetry.

Kurtosis — Kurtosis measures the heaviness of the tails of a distribution. It helps identify whether a distribution contains more extreme values than a normal distribution.

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9. Interactive Filters

The dashboard contains the following sidebar filters:

- Age range
- BMI range
- Smoking status
- Region

The filtered dataset is created using Boolean conditions:

```
filtered_df = df[
    (df["age"] >= age_range[0])
    & (df["age"] <= age_range[1])
    & (df["bmi"] >= bmi_range[0])
    & (df["bmi"] <= bmi_range[1])
    & (df["smoker"].isin(selected_smokers))
    & (df["region"].isin(selected_regions))
]
```

The filters allow the user to analyze a selected subpopulation. When the filter values change, the dashboard updates: Number of observations, Mean charges, Median charges, Mean BMI, Descriptive statistics, Histograms, Scatter plots, Correlation matrix.

If no observations match the selected filters, the dashboard displays a warning message.

The two-group hypothesis test and the one-way ANOVA use the complete, unfiltered dataset rather than the sidebar-filtered dataset. Sidebar filters affect the exploratory data analysis, the regression model, and the residual diagnostics, but they do not change the hypothesis-test results.

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10. Summary Metrics

The dashboard displays four summary metrics:

1. Number of observations.
2. Mean medical charges.
3. Median medical charges.
4. Mean BMI.

These metrics provide a quick overview of the currently selected data.

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11. Visual Exploration

11.1 Distribution of Medical Charges — A histogram with a KDE curve is used to display the distribution of medical charges. The histogram shows how frequently different charge values occur, and the KDE curve provides a smooth estimate of the distribution. Medical charges are positively skewed because a smaller number of individuals have very high charges.

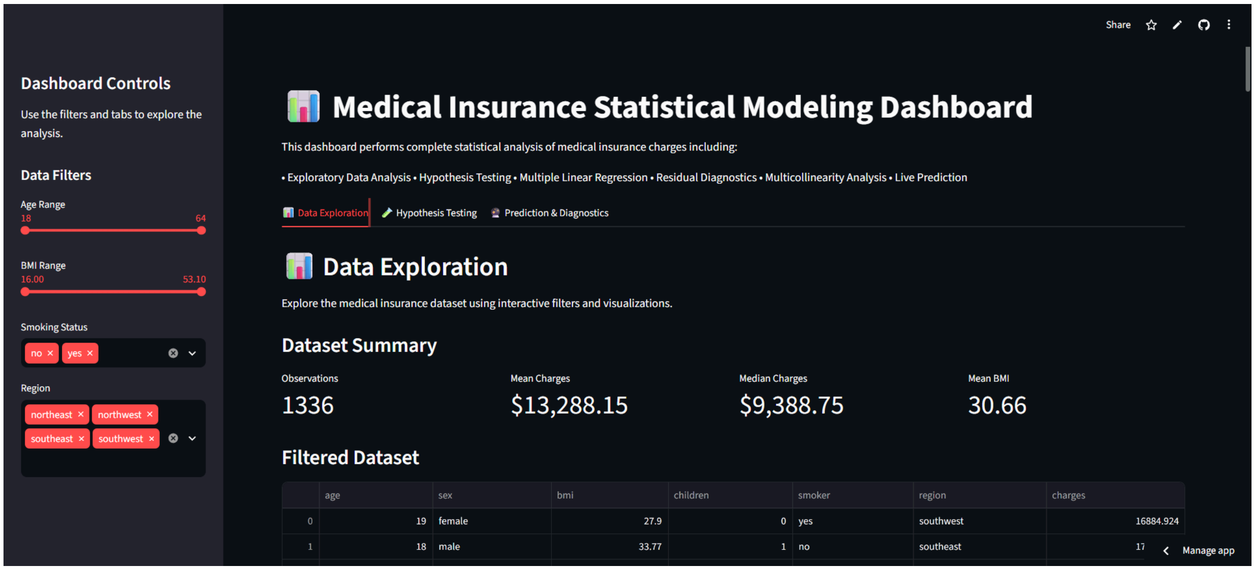
11.2 Age versus Medical Charges — A scatter plot is used to display the relationship between age and medical charges. The points are colored according to smoking status, making it easier to observe whether smokers and non-smokers have different charge patterns.

11.3 BMI versus Medical Charges — A scatter plot is used to examine the relationship between BMI and medical charges. The smoking-status color grouping helps identify whether BMI has a different effect among smokers and non-smokers.

11.4 Correlation Matrix — A correlation matrix is created for age, BMI, children, and charges. Correlation measures the strength and direction of a linear relationship, and correlation values range between -1 and +1:

- +1 indicates a perfect positive relationship.
- -1 indicates a perfect negative relationship.
- 0 indicates no linear relationship.

A Seaborn heatmap is used to display the correlation values.



PART B: HYPOTHESIS TESTING

12. Significance Level

α = 0.05

Decision rule:

- If p-value < 0.05, reject the null hypothesis.
- If p-value ≥ 0.05, fail to reject the null hypothesis.

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13. Hypothesis Test 1: Two-Group Comparison

The default comparison is between smokers and non-smokers, using the numerical variable charges.

Null Hypothesis: There is no significant difference in medical charges between smokers and non-smokers.

Alternative Hypothesis: There is a significant difference in medical charges between smokers and non-smokers.

The dashboard also allows the user to select a categorical grouping variable, a numerical variable, and exactly two groups.

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14. Shapiro-Wilk Normality Test

The Shapiro-Wilk test is performed separately for each group to check whether the observations follow a normal distribution.

Null Hypothesis: The observations follow a normal distribution.

Alternative Hypothesis: The observations do not follow a normal distribution.

For this dataset, the Shapiro-Wilk p-values for both smokers and non-smokers are below 0.05. Therefore, the null hypothesis of normality is rejected.

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15. Levene's Test

Levene's test checks whether the variances of two groups are equal.

Null Hypothesis: The variances of the two groups are equal.

Alternative Hypothesis: The variances of the two groups are significantly different.

The p-value of Levene's test is below 0.05. Therefore, the equal-variance assumption is rejected.

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16. Automatic Statistical Test Selection

The application automatically selects the appropriate test. If both groups are normally distributed, an independent two-sample t-test is selected:

```
stats.ttest_ind()
```

If at least one group is not normally distributed, the Mann-Whitney U test is selected:

```
stats.mannwhitneyu()
```

Since both medical charge groups are non-normal, the application selects the Mann-Whitney U test.

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17. Results of the Two-Group Test

Group	Mean Charges
Smokers	$32,050.23
Non-smokers	$8,434.27

Statistic	Value
U statistic	≈ 284,133
p-value	< 0.001

Since the p-value is less than 0.05, the null hypothesis is rejected.

Conclusion: There is statistically significant evidence that medical charges differ between smokers and non-smokers. The average charges for smokers are substantially higher than the average charges for non-smokers.

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18. Hypothesis Test 2: One-Way ANOVA

The second hypothesis test compares medical charges across regions: Northeast, Northwest, Southeast, and Southwest.

Null Hypothesis: The mean medical charges are equal across all regions.

Alternative Hypothesis: At least one regional mean is different.

The dashboard also allows the user to choose the grouping variable and numerical variable dynamically.

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19. ANOVA Method

The dataset is divided into separate groups according to region. The medical charges for each region are stored in separate groups and passed to:

stats.f_oneway()

ANOVA compares the variation between group means with the variation within groups:

F-statistic = Between-group variation / Within-group variation

A larger F-statistic indicates greater evidence that the group means are different.

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20. ANOVA Results

Region	Mean Charges
Northeast	$13,406.38
Northwest	$12,417.58
Southeast	$14,735.41
Southwest	$12,346.94

Statistic	Value
F-statistic	≈ 2.970
p-value	≈ 0.0309

Since the p-value is below 0.05, the null hypothesis is rejected.

Conclusion: At least one region has a statistically different mean medical charge. ANOVA confirms that a significant difference exists, but it does not identify which specific pairs of regions differ. A post-hoc test such as Tukey's HSD would be required for pairwise comparisons.

PART C: MULTIPLE LINEAR REGRESSION

21. Regression Model

The project uses multiple linear regression to estimate medical insurance charges. The general regression model is:

$Y = \beta 0 + \beta 1X1 + \beta 2X2 + ... + \beta kXk + \varepsilon$

Where Y is the dependent variable, β0 is the intercept, β1 through βk are regression coefficients, X1 through Xk are predictor variables, and ε is the error term.

In this project, Y = charges, and the predictors are: Age, BMI, Number of children, Sex, Smoking status, Region.

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22. Categorical Variable Encoding

OLS regression requires numerical predictor variables. The categorical variables are converted into dummy variables using:

pd.get_dummies(

```
    df[predictor_cols],
    drop_first=True,
    dtype=float
)
```

The `drop_first=True` argument removes one category from each categorical variable. This avoids perfect multicollinearity, also known as the dummy variable trap.

For example, for the sex variable, Female becomes the reference category and Male is represented using `sex_male`. The coefficient of `sex_male` is interpreted relative to the female reference group.

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23. Adding the Intercept

```
X = sm.add_constant(X)
```

The intercept represents the expected value of the response when all numerical predictors are zero and all categorical variables are at their reference levels.

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24. Fitting the OLS Model

```
model = sm.OLS(y, X).fit()
```

The fitted model provides: Coefficients, Standard errors, t-statistics, p-values, Confidence intervals, R-squared, Adjusted R-squared, F-statistic.

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25. Regression Coefficients

A regression coefficient measures the expected change in charges for a one-unit change in a predictor while holding all other variables constant. Examples:

- The age coefficient represents the expected change in charges for a one-year increase in age.
- The BMI coefficient represents the expected change in charges for a one-unit increase in BMI.
- The smoker coefficient compares smokers with non-smokers.

- The region coefficients compare each region with the reference region.

Predictor	**Coefficient**	**p-value**	**95% CI**
Intercept	≈ -11,938.54	< 0.001	[-13,879.98, -9,997.10]
age	≈ 256.86	< 0.001	[229.44, 284.28]
bmi	≈ 339.19	< 0.001	[290.89, 387.50]
children	≈ 475.50	0.017	[85.68, 865.32]
sex_male	≈ -131.31	0.694	[-789.05, 526.43]
smoker_yes	≈ 23,848.53	< 0.001	[23,296.68, 24,400.37]
region_northwest	≈ -352.96	0.443	[-1,258.09, 552.18]
region_southeast	≈ -1,035.02	0.028	[-1,959.20, -110.85]
region_southwest	≈ -960.05	0.043	[-1,884.63, -35.46]

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26. <u>P-values</u>

The p-value tests whether a regression coefficient is statistically different from zero.

Null Hypothesis: The coefficient is equal to zero.

Alternative Hypothesis: The coefficient is not equal to zero.

A p-value below 0.05 indicates that the predictor has a statistically significant relationship with charges after controlling for the other variables.

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27. <u>Confidence Intervals</u>

The project calculates 95% confidence intervals for every coefficient. A confidence interval represents a range of plausible values for the true population coefficient. If the confidence interval does not include zero, the coefficient is generally considered statistically significant at the 5% level.

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28. Model Fit

Metric	Value
R-squared	≈ 0.7509
Adjusted R-squared	≈ 0.7494

R-squared — The R-squared value represents the proportion of variation in charges explained by the predictors. The model explains approximately 75.09% of the variation in medical charges.

Adjusted R-squared — Adjusted R-squared accounts for the number of predictors in the model. It is useful because it penalizes the addition of variables that do not improve the model significantly.

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29. Filter-Based Regression

The dashboard uses the filtered dataset for regression and diagnostic analysis when at least 30 observations remain. If fewer than 30 observations remain after applying the filters, the complete dataset is used to avoid unstable statistical estimates.

This means that changing the sidebar filters can change: Regression coefficients, P-values, R-squared, Fitted values, Residuals, Q-Q plot, Jarque-Bera results, VIF values.

PART D: LIVE PREDICTION

30. User Input Controls

The prediction section accepts the following user inputs: Age, BMI, Number of children, Sex, Smoking status, Region.

These values are collected into a one-row DataFrame.

---

31. Preparing Prediction Data

The prediction input is encoded using the same dummy-variable method used during model training. The columns are then reindexed to match the training columns:

```
input_encoded = input_encoded.reindex(
    columns=model_data.columns,
    fill_value=0
)
```

This is important because the prediction DataFrame must have exactly the same variables and column ordering as the model training DataFrame.

---

32. Generating Predictions

```
prediction = model.get_prediction(input_encoded)
```

The dashboard displays:

1. Predicted medical charges.
2. 95% confidence interval.
3. 95% prediction interval.

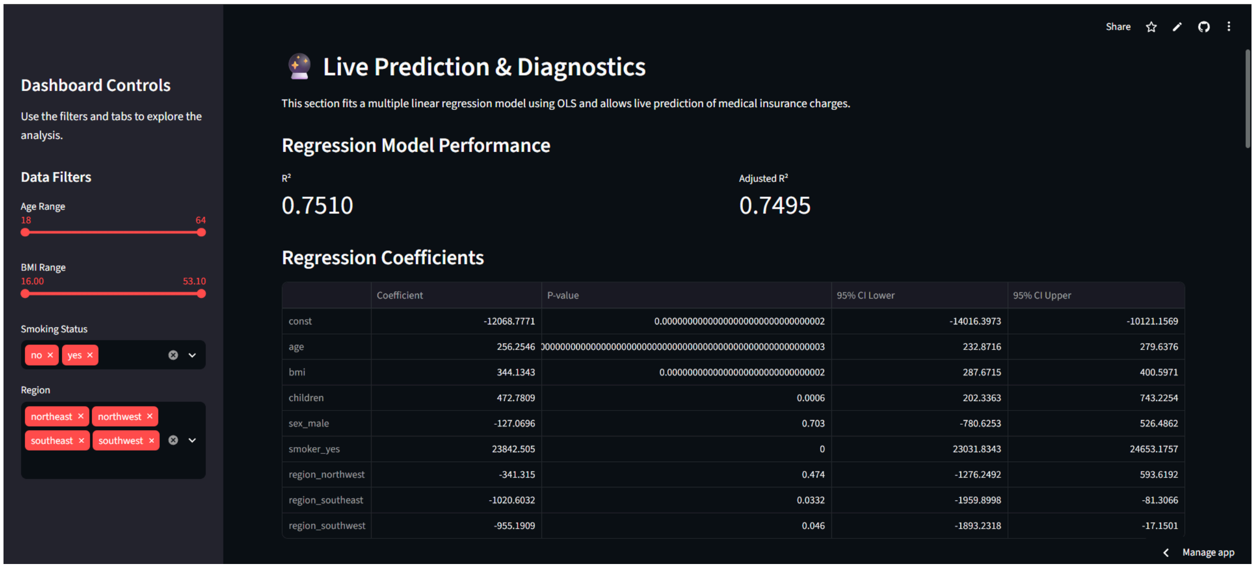
---

33. Confidence Interval and Prediction Interval

A confidence interval estimates the uncertainty around the expected mean charge for people with a given set of characteristics.

A prediction interval estimates the range in which the charge for one future individual may fall. The prediction interval is wider because it includes:

- Uncertainty in the estimated mean.
- Individual-level variation.
- Random error.



PART E: RESIDUAL DIAGNOSTICS

34. Residuals

A residual is the difference between the observed value and the fitted value:

Residual = Observed charge - Predicted charge

```
fitted_values = model.fittedvalues
residuals = model.resid
```

Residual analysis is important for checking whether the assumptions of linear regression are reasonable.

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35. Residuals versus Fitted Values

The residuals-versus-fitted plot is used to investigate: Linearity, Homoscedasticity, Outliers, Systematic patterns.

Ideally, the residuals should be randomly scattered and centered around zero, and the spread should remain approximately constant. A curved pattern may indicate non-linearity, and a funnel-shaped pattern may indicate heteroscedasticity.

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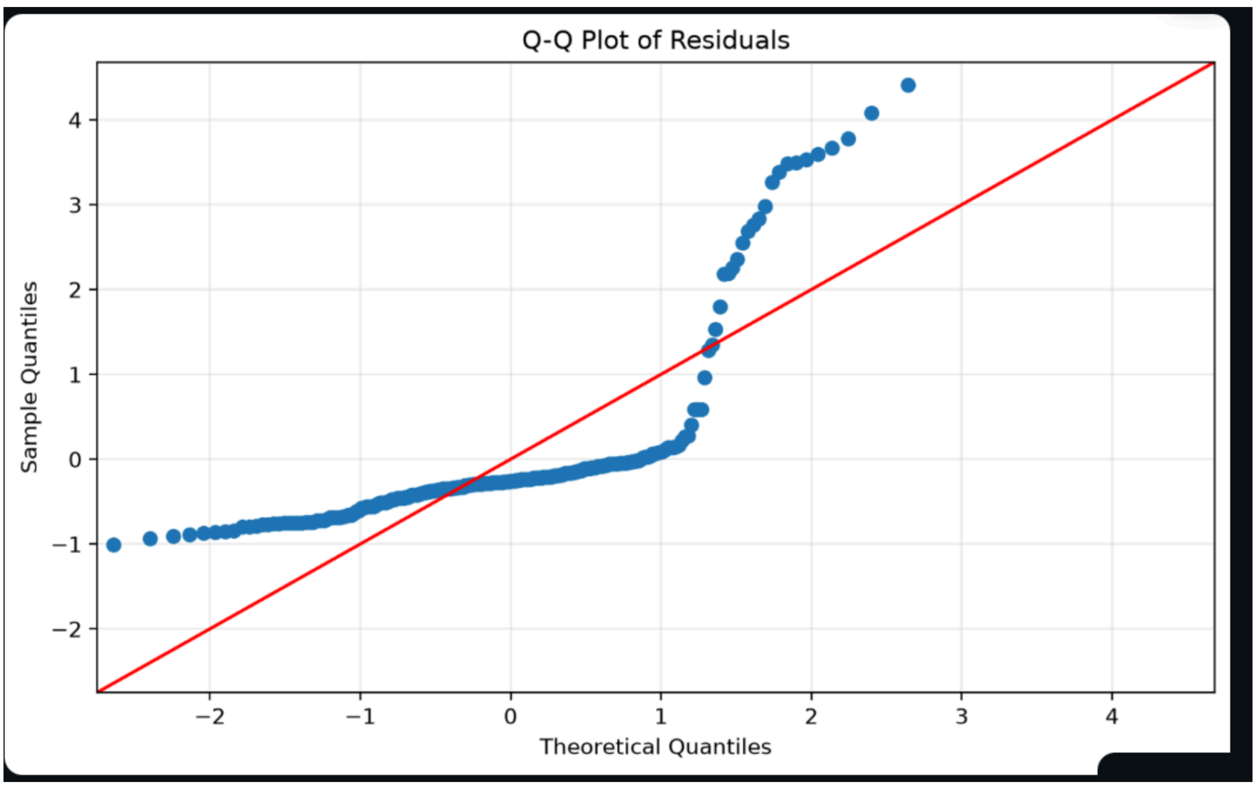
36. Q-Q Plot

The Q-Q plot compares the residual quantiles with theoretical normal quantiles.

```
sm.qqplot(
    residuals,
    line="45",
    fit=True,
    ax=ax
)
```

If the residuals are approximately normally distributed, the points should lie close to the diagonal reference line. Large deviations from the diagonal indicate that the residuals are not normally distributed. The Q-Q plot is generated after the OLS model has been fitted and residuals have been calculated.



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37. Jarque-Bera Test

The Jarque-Bera test checks whether residuals have the skewness and kurtosis expected under normality.

Null Hypothesis: The residuals are normally distributed.

Alternative Hypothesis: The residuals are not normally distributed.

Statistic	Value
Jarque-Bera statistic	718.887
p-value	< 0.001

Since the p-value is below 0.05, the null hypothesis is rejected.

Conclusion: The residuals show statistically significant evidence of departure from normality.

Residual normality is mainly important for reliable small-sample hypothesis tests and confidence intervals. The OLS coefficient estimates can still be calculated when residuals are non-normal. Given the large sample size (n = 1,338), this is treated as a limitation rather than a fatal flaw of the model.

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38. Variance Inflation Factor

VIF is used to detect multicollinearity. VIF values are calculated for Age, BMI, and Children.

VIF Value	Interpretation
Approximately 1	Very little multicollinearity
1 to 5	Usually acceptable
Greater than 5	Potential concern
Greater than 10	Serious concern

Variable	Approximate VIF
Age	1.014

BMI	1.012
Children	1.002

All values are close to 1.

Conclusion: There is very little evidence of multicollinearity among the continuous predictors.

PART F: OPTIONAL INTERACTION MODEL

39. BMI and Smoking Interaction

The standalone analysis also includes an optional interaction model between BMI and smoking status. First, smoking status is converted into a numerical binary variable:

```
df["smoker_num"] = (
    df["smoker"] == "yes"
).astype(int)
```

The interaction variable is then created:

```
df["bmi_smoker"] = (
    df["bmi"] * df["smoker_num"]
)
```

The interaction term measures whether the effect of BMI on medical charges differs between smokers and non-smokers. The interaction coefficient is approximately 1433.79 and its p-value is extremely small.

Interpretation: The relationship between BMI and medical charges is significantly different for smokers and non-smokers. This indicates that smoking status modifies the effect of BMI on insurance charges.

PART G: MAIN FINDINGS

40. Important Statistical Findings

1. The dataset contains 1,338 observations and 7 variables.
2. There are no missing values.
3. There is one duplicate row.
4. Smokers have much higher average charges than non-smokers.
5. Mean smoker charges are approximately $32,050.23.
6. Mean non-smoker charges are approximately $8,434.27.
7. Both charge groups fail the normality test.
8. The variances of the groups are significantly different.
9. The Mann-Whitney U test confirms a significant difference in charges.
10. Regional charge means are significantly different according to ANOVA.
11. The regression model explains approximately 75% of the variation in charges.
12. Age, BMI, number of children, and smoking status are statistically significant predictors.
13. Sex is not statistically significant in the fitted model.
14. The continuous predictors have very low VIF values.
15. The residuals are not normally distributed.
16. The BMI-smoking interaction is statistically significant.

PART H: DEPLOYMENT

41. Local Execution

Create a virtual environment:

python -m venv venv

Activate the virtual environment:

.\venv\Scripts\Activate.ps1

Install the required packages:

python -m pip install -r requirements.txt

Run the dashboard:

.\venv\Scripts\streamlit.exe run app.py

The application opens at:

http://localhost:8501

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42. Streamlit Cloud Deployment

The dashboard is deployed through Streamlit Community Cloud.

Hosted link: https://medical-insurance-statistical-modeling-dashboard.streamlit.app/

The main Streamlit file is:

202618052_Lab_4/app.py

The repository must contain: `app.py`, `requirements.txt`, `data/insurance.csv`

The root-level `requirements.txt` file contains the packages required by the dashboard.

PART I: LIMITATIONS

43. Project Limitations

1. The dataset is observational, so statistical relationships do not prove causation.
2. The regression model assumes linear relationships between predictors and charges.
3. The residual normality assumption is not satisfied.
4. ANOVA only confirms that at least one group differs; it does not identify which groups differ.
5. A Tukey HSD post-hoc test could be added for pairwise regional comparisons.
6. Medical charges may contain influential outliers.
7. VIF was calculated only for continuous predictors.
8. The model does not include all possible medical, economic, or lifestyle factors.
9. Prediction results depend on the quality and limitations of the dataset.
10. Statistical results may change when different filters are applied.

44. Conclusion

This project demonstrates a complete statistical modeling workflow using a real-world medical insurance dataset. The project includes: Data loading and validation, Descriptive statistics, Interactive filtering, Distribution plots, Scatter plots, Correlation analysis, Shapiro-Wilk testing, Levene's testing, Mann-Whitney U testing, One-way ANOVA, Multiple linear regression, Coefficient interpretation, Confidence intervals, Live predictions, Prediction intervals, Residual diagnostics, Q-Q plots, Jarque-Bera testing, Variance Inflation Factor analysis, Interaction modeling, Streamlit dashboard development, Cloud deployment.

The analysis shows that smoking status is the strongest practical factor associated with medical insurance charges. Age, BMI, and the number of children also contribute to the prediction of charges.

The dashboard makes the statistical analysis interactive and allows users to explore how different subsets of the data affect descriptive results, model estimates, predictions, and diagnostic conclusions.